

(192511470)

SET - 4

QUESTION 1: LEFT JOIN

```
mysql> SELECT
->     c.course_id,
->     c.course_name,
->     c.credits,
->     i.name AS instructor_name
-> FROM Courses c
-> LEFT JOIN Instructors i
-> ON c.instructor_id = i.instructor_id;
```

course_id	course_name	credits	instructor_name
101	Database Management Systems	4	Dr. Kumar
102	Computer Networks	3	Dr. Priya
103	Operating Systems	4	Dr. Ravi
104	Artificial Intelligence	3	NULL

4 rows in set (0.00 sec)

QUESTION 2: VIEW

```
mysql> CREATE VIEW vw_dean_list AS
-> SELECT
->     student_id,
->     CONCAT(first_name, ' ', last_name) AS full_name,
->     major,
->     gpa
-> FROM Students
-> WHERE gpa >= 3.5;
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> SELECT * FROM vw_dean_list;
```

student_id	full_name	major	gpa
1	Arun Kumar	CSE	3.80
2	Priya Raj	ECE	3.60
4	Santhosh E	CSE	3.90

3 rows in set (0.00 sec)

QUESTION 3: FUNCTION

```

mysql> DELIMITER //
mysql> CREATE FUNCTION fn_calculate_letter_grade(mark INT)
  -> RETURNS CHAR(1)
  -> DETERMINISTIC
  -> BEGIN
  ->     IF mark >= 90 AND mark <= 100 THEN
  ->         RETURN 'A';
  ->     ELSEIF mark >= 80 THEN
  ->         RETURN 'B';
  ->     ELSEIF mark >= 70 THEN
  ->         RETURN 'C';
  ->     ELSEIF mark >= 60 THEN
  ->         RETURN 'D';
  ->     ELSE
  ->         RETURN 'F';
  ->     END IF;
  -> END //
Query OK, 0 rows affected (0.07 sec)

mysql> DELIMITER ;
mysql> SELECT fn_calculate_letter_grade(95);
+-----+
| fn_calculate_letter_grade(95) |
+-----+
| A                               |
+-----+
1 row in set (0.04 sec)

```

QUESTION 4: TRIGGER

```

mysql> DELIMITER //
mysql> CREATE TRIGGER trg_check_gpa
  -> BEFORE INSERT ON Students
  -> FOR EACH ROW
  -> BEGIN
  ->     IF NEW.gpa < 0.0 OR NEW.gpa > 4.0 THEN
  ->         SIGNAL SQLSTATE '45000'
  ->         SET MESSAGE_TEXT =
  ->         'GPA must be between 0.0 and 4.0';
  ->     END IF;
  -> END //
Query OK, 0 rows affected (0.05 sec)

```